

Name: _____ Date: _____

These are practice questions that I have created to prepare me and other students for our **quiz** on September 14, 2026, for **MTH 181-03: Discrete Mathematics I** class.

Set 1.1

1. What type of mathematical statement is this?

”Is there a real number whose square is two?”

A. Universal B. Conditional C. Existential

2. Rewrite the following statement **less formally**, without using variables. Also determine whether the statement is TRUE or FALSE.

There are integers a and b , where $\sqrt{a^2 + b^2} \geq a + b$.

3. Explain the difference, please.

(a) Universal Statement:

(b) Conditional Statement:

(c) Existential Statement:

Set 1.2

4. The axiom of extension states that the order of elements in a set _____.

5. Answer the follow three subparts:

(a) $\mathbb{R}$ = The set of all _____.

(b) $\mathbb{Z}$ = The set of all _____.

(c) $\mathbb{Q}$ = The set of all _____.

6. Lets $A = \{2, 4, 6\}$ and $B = \{4, 2, 4, 6, 6\}$.

Is $A \neq B$? Select one, please.

☐ True

☐ False

7. Set-Builder Notation to Set-Roster Notation

$\{x \in \mathbb{Z}^+ \mid 2 \leq x < 7\}$

8. Is $\{6\} \in \{1, 6, 3\}$? Please, select one.

☐ True

☐ False

9. Is $7 \subseteq \{7\}$ Please, select one.

☐ True

☐ False

10. Let $A = \{x, y, z\}$ and $B = \{1, 2, 3\}$.

$A \times B =$

11. Let $A = \{1, 2, 3\}$, $B = \{x\}$, and $C = \{y, z\}$.

$A \times (B \times C) =$

12. Let $A = \emptyset$ and $B = \{1, 2, 3\}$

$A \times B =$

13. Strings: Let $A = \{a, b\}$. List all the strings of length four over A with at least three characters that are the same.

14. Let $A = \{1, 2, 3\}$ and $B = \{1, 3, 3, 2, 4\}$. Is A a proper subset of B ?

☐ True

☐ False